

SSD_pure version LINUX configuration reference

1. Disable mi and mhal modules in

project\release\customer_tailor\nvr_i2m_display_glibc_tailor.mk:

```
include $(PROJ_ROOT)/release/customer_tailor/nvr_default.mk
interface_ai:=disable
interface_ao:=disable
interface_gfx:=disable
interface_hdmi:=disable
interface_divp:=disable
interface_disp:=disable
interface_panel:=disable
interface_rgn:=disable
interface_shadow:=disable
interface_uac:=disable
interface_vdf:=disable
interface_vdisp:=disable
interface_vdec:=disable
interface_venc:=disable
interface_wlan:=disable

misc_fbdev:=enable
#verify_zk_full:=enable
#mhal
mhal_aio:=disable
mhal_csi:=disable
mhal_disp:=disable
mhal_divp:=disable
mhal_isp:=disable
mhal_ispalgo:=disable
mhal_ispmid:=disable
mhal_ldc:=disable
mhal_mload:=disable
mhal_panel:=disable
mhal_rgn:=disable
mhal_sensorif:=disable
mhal_venc:=disable
mhal_vif:=disable
mhal_vpe:=disable
mhal_hdmidx:=disable
```

2. project\release\nvr\i2m\common\glibc\8.2.1\modules\4.9.84\modules_depend
delete all mi modules
3. project\release\nvr\i2m\common\glibc\8.2.1\modules\4.9.84\misc_mod_list
delete the mhal module
4. project\release\nvr\i2m\common\glibc\8.2.1\modules\4.9.84\misc_mod_list_late
delete the fbdev module
5. In project\image\configs\i2m\rootfs.mk, delete the following red box mi part:

```

115 echo "#mi module" >> $(OUTPUTDIR)/customer/demo.sh; \
116 fi;
117
118 if [ -f "$(LIB_DIR_PATH)/modules/$(KERNEL_VERSION)/misc_mod_list_late" ]; then \
119 cat $(LIB_DIR_PATH)/modules/$(KERNEL_VERSION)/misc_mod_list_late | sed 's/(\.*)ko$insmod /config/modules/$(KERNEL_VERSION)/\1.ko$' >> $(OUTPUTDIR)
120 cat $(LIB_DIR_PATH)/modules/$(KERNEL_VERSION)/misc_mod_list_late | sed 's/(\.*)ko\(\.*)$(LIB_DIR_PATH)/modules/$(KERNEL_VERSION)/\1.ko$' | \
121 xargs -i cp -rvf {} $(miservice$(RESOURCE))/modules/$(KERNEL_VERSION); \
122 echo "#misc_mod_list_late" >> $(OUTPUTDIR)/customer/demo.sh; \
123 fi;
124
125 #sed -i 's/mi_common/insmod /config/modules/$(KERNEL_VERSION)/mi_common.ko$major='cat /proc/devices | busybox awk "\\\$2==\\\"mi\\\" {print .
126 #sed -i 's/#mi module/$major='cat /proc/devices | busybox awk "\\\$2==\\\"mi_poll\\\" {print \\\$1}\\\"nbusybox mknod /dev/mi_poll c $major 0' $
127 if [ $(PHY_TEST) = "yes" ]; then \
128 echo -e "\033[41;33;5m !!! Replace \"mdrv-sata-host.ko\" with \"sata_bench_test.ko\" !!!\033[0m" ; \
129 sed '/mdrv-sata-host/d' $(OUTPUTDIR)/customer/demo.sh > $(miservice$(RESOURCE))/demotemp.sh ; \
130 echo insmod /config/sata_phy/sata_bench_test.ko >> $(miservice$(RESOURCE))/demotemp.sh ; \
131 cp $(miservice$(RESOURCE))/demotemp.sh $(OUTPUTDIR)/customer/demo.sh ; \
132 rm $(miservice$(RESOURCE))/demotemp.sh ; \
133 fi;

```

6. In project\image\configs\i2m\rootfs.mk, delete the following sdk-related bins in the red box:

```

.PHONY: rootfs root app
rootfs:root app
root:
cd rootfs; tar xf rootfs.tar.gz -C $(OUTPUTDIR)
tar xf busybox/$(BUSYBOX).tar.gz -C $(OUTPUTDIR)/rootfs
tar xf $(LIB_DIR_PATH)/package/$(LIBC).tar.gz -C $(OUTPUTDIR)/rootfs/lib
cp $(LIB_DIR_PATH)/mi_libs/dynamic/* $(OUTPUTDIR)/rootfs/lib/
cp $(LIB_DIR_PATH)/ex_libs/dynamic/* $(OUTPUTDIR)/rootfs/lib/

mkdir -p $(miservice$(RESOURCE))

if [ "$(BOARD)" = "010A" ]; then \
cp -rf $(PROJ_ROOT)/board/ini/* $(miservice$(RESOURCE)) ; \
else \
cp -rf $(PROJ_ROOT)/board/ini/LCM $(miservice$(RESOURCE)) ; \
fi;

if [ "$(BOARD)" = "010A" ]; then \
cp -rf $(PROJ_ROOT)/board/$(CHIP)/$(BOARD_NAME)/config/* $(miservice$(RESOURCE)) ; \
else \
cp -rf $(PROJ_ROOT)/board/$(CHIP)/$(BOARD_NAME)/config/fbdev.ini $(miservice$(RESOURCE)) ; \
fi;

cp -vf $(PROJ_ROOT)/board/$(CHIP)/mmap/$(MMAP) $(miservice$(RESOURCE))/mmap.ini
cp -rvf $(LIB_DIR_PATH)/bin/config_tool/* $(miservice$(RESOURCE))
cd $(miservice$(RESOURCE)); chmod +x config_tool; ln -sf config_tool dump_config; ln -sf config_tool dump_mmap
if [ "$(BOARD)" = "010A" ]; then \
cp -rf $(PROJ_ROOT)/board/$(CHIP)/pq $(miservice$(RESOURCE)) ; \
find $(miservice$(RESOURCE))/pq/ -type f ! -name "Main.bin" -type f ! -name "Main_Ek.bin" -type f ! -name "Bandwidth_BegTable.bin" | xargs rm -rf ; \
fi;

if [ $(interface_vdec) = "enable" ]; then \
cp -rf $(PROJ_ROOT)/board/$(CHIP)/vdec_fw $(miservice$(RESOURCE)); \
fi;

mkdir -p $(OUTPUTDIR)/rootfs/config
cp -rf etc/* $(OUTPUTDIR)/rootfs/etc

```

7. In project\image\configs\i2m\rootfs.mk, delete the following bins related to sdk debug in the red box:

```

cp -rvf $(PROJ_ROOT)/tools/$(TOOLCHAIN)/$(TOOLCHAIN_VERSION)/fw_printenv/* $(OUTPUTDIR)/rootfs/etc/
echo "#(ENV_CFG)" > $(OUTPUTDIR)/rootfs/etc/fw_env.config
cd $(OUTPUTDIR)/rootfs/etc; ln -sf fw_printenv fw_setenv
echo mkdir -p /var/lock >> $(OUTPUTDIR)/rootfs/etc/profile
if [ "$(FPGA)" = "1" ]; then \
echo mount -t jffs2 /dev/mtdblock1 /config >> $(OUTPUTDIR)/rootfs/etc/profile ; \
fi;
echo -e $(foreach block, $(USR_MOUNT_BLOCKS), "mount -t ${block} ${FSTYPE} ${block} ${MOUNTPT} ${block} ${MOUNTG})\n
chmod 755 $(LIB_DIR_PATH)/bin/debug/*
cp -rf $(LIB_DIR_PATH)/bin/debug/* $(miservice$(RESOURCE))
rm -rf $(OUTPUTDIR)/customer/demo.sh
mkdir -p $(OUTPUTDIR)/customer
touch $(OUTPUTDIR)/customer/demo.sh
chmod 755 $(OUTPUTDIR)/customer/demo.sh

```

8. The config sz size used by the project is set to 0:

```

TOOLCHAIN_REL = arm-linux-gnueabi
KERNEL_VERSION = 4.9.84
LIBC = libc-2.28
BUSYBOX = busybox-1.20.2-arm-linux-gnueabi-hf-glibc-8.2.1-dynamic
KERNEL_CONFIG = glibc
IMAGE_CONFIG = spinand.ubifs.p2.partition.config
CUSTOMER_OPTIONS = 011a.201_options.mk
CUSTOMER_TAILOR = nvr_i2m_display_glibc_tailor.mk
MMAP = MMAP_I2M_64M.h
MHAL = i2m
MERGE_BOOT = TRUE
BOOTLOGO_FILE = sigmastar1024_600.jpg
#BOOTLOGO_ADDR = E_LX_FB
LOGO_ADDR = 0x3D00000
BOOTLOGO_BUFSIZE = 0x200000
PANEL_NAME = SAT070CP50
PANEL_WIDTH = 1024
PANEL_HEIGHT = 600
EXBOOTARGS =
KERNEL_BOOT_ENV = LX_MEM=$(KERNEL_MEMLN) mma_heap=mma_heap_name0,miu=0,sz=0x00000 mma_memblock_remove=1
TOOLCHAIN_REL = arm-linux-gnueabi-hf-

```

After cutting, the memory of linux is:

```

on (31) / # cat /proc/meminfo
MemTotal:      59028 kB
MemFree:       47292 kB
MemAvailable:  46592 kB
Buffers:       0 kB
Cached:       980 kB
SwapCached:    0 kB

```

The distribution of DDR is as follows:

64M DDR ~ = Total(59028KB) + kernel text(4644KB) + MMAP(1024KB)

kernel text:

```

/ # cat /sys/kernel/debug/memblock/reserved
0: 0x20004000..0x20007fff
1: 0x20008240..0x204912cf
2: 0x23c00000..0x23dfffff
3: 0x23e31000..0x23e58fff
4: 0x23e5b00c..0x23efefff
5: 0x23eff440..0x23eff87b

```

kernel used = Total(59028KB) - MemFree(47292KB) = 11736KB

